

2.5

```
1  #include <iostream>
2  #include <stdlib.h>
3  using namespace std;
4
5  class Student
6  {
7  private:
8      int marks[5];
9
10 public:
11     friend void calculateAverage(Student s);
12     void input()
13     {
14         int i = 0;
15         while (i < 5)
16         {
17             cout << "Enter Sub " << i + 1 << " Marks(0-100): ";
18             cin >> marks[i];
19             if (marks[i] >= 0 && marks[i] <= 100)
20             {
21                 i++;
22             }
23         }
24     }
25 };
26
27 void calculateAverage(Student s)
28 {
29     float sum = 0;
30     for (int i = 0; i < 5; i++)
31     {
32         sum += s.marks[i];
33     }
34     cout << "Average : " << sum / 5 << endl;
35 }
36
37 int main()
38 {
39     system("color 04");
40     Student s1;
41     s1.input();
42     calculateAverage(s1);
43 }
```

```
Enter Sub 1 Marks(0-100): 25
Enter Sub 2 Marks(0-100): 25
Enter Sub 3 Marks(0-100): 25
Enter Sub 4 Marks(0-100): 25
Enter Sub 5 Marks(0-100): 25
Average : 25
```

3.6

```
1  #include <iostream>
2  using namespace std;
3  class LibraryBook
4  {
5  private:
6      string title;
7      string author;
8      bool isIssue;
9
10 public:
11     LibraryBook(string t, string a)
12     {
13         title = t;
14         author = a;
15         isIssue = false;
16     }
17     friend class Librarian;
18 };
```

```
19 class Librarian
20 {
21 public:
22
23     void Issue(LibraryBook &b)
24     {
25         if (b.isIssue == false)
26         {
27             b.isIssue = true;
28             cout << "Book Issued" << endl;
29         }else{
30             cout<<"Book is Already Issued"<<endl;
31         }
32     }
33
34     void ShowBook(LibraryBook b){
35         cout<<"Title: "<<b.title<<" Author: "<<b.author<<endl;
36     }
37
38 };
39
40
```

```
41 int main()
42 {
43     LibraryBook b1("Hero", "Yash");
44     Librarian l1;
45     l1.Issue(b1);
46     l1.ShowBook(b1);
47     Librarian l2;
48     l2.Issue(b1);
49     LibraryBook b2("Animal Farm", "George Orwell");
50     l2.Issue(b2);
51     l2.ShowBook(b2);
52
53 }
```

```
Book Issued
Title: Hero Author: Yash
Book is Already Issued
Book Issued
Title: Animal Farm Author: George Orwell
```

Task-1

```
1  #include<iostream>
2  using namespace std;
3  class Rectangle{
4  private:
5      double length;
6      double width;
7  public:
8      void GetInput(){
9          while(true){
10             cout<<"Enter Length: ";
11             cin>>length;
12             cout<<"Enter Width: ";
13             cin>>width;
14             if(width >=0 && length >=0){
15                 break;
16             }else{
17                 cout<<"These Values Can't be Negative!"<<endl;
18             }
19         }
20     }
21     friend void Perimeter(Rectangle r1);
22     friend void Area(Rectangle r1);
23 };
24 void Perimeter(Rectangle r1){
25     cout<<"Perimeter: "<<(2*r1.length)+(2*r1.width)<<endl;
26 }
27 void Area(Rectangle r1){
28     cout<<"Area: "<<r1.length*r1.width<<endl;
29 }
```

```
01 int main(){
02     Rectangle r;
03     r.GetInput();
04     Perimeter(r);
05     Area(r);
06 }
```

```
Enter Length: 25
Enter Width: 25
Perimeter: 100
Area: 625
```

Task-2

```
1  #include<iostream>
2  using namespace std;
3  class Mechanic;
4  class CarEngine{
5  private:
6      int horseP;
7  public:
8      CarEngine(int hp){
9          horseP=hp;
10     }
11     friend class Mechanic;
12 };
13 class Mechanic{
14 public:
15     void DisplayHP(CarEngine engine){
16         cout<<"Horse Power: "<<engine.horseP<<endl;
17     }
18     void UpgradeHP( CarEngine &engine){
19         while(true){
20             cout<<"Enter HorsePower(100-1000): ";
21             cin>>engine.horseP;
22             if(engine.horseP > 100 && engine.horseP <1000){
23                 break;
24             }else{
25                 cout<<"Must Be Between 100-1000!"<<endl;
26             }
27         }
28     }
29 };
30
```

```
30 int main(){
31     CarEngine engine(490);
32     Mechanic m;
33     m.DisplayHP(engine);
34     m.UpgradeHP(engine);
35     m.DisplayHP(engine);
36 }
```

```
Horse Power: 490
Enter HorsePower(100-1000): 25
Must Be Between 100-1000!
Enter HorsePower(100-1000): 650
Horse Power: 650
```

Task-3 and 4.6

```
1  #include<iostream>
2  using namespace std;
3  class Person{
4      protected:
5          int id;
6          string name;
7      public:
8
9          Person(int id,string n):id(id),name(n){}
10     void displayPerson(){
11         cout<<"Id: "<<id<<endl<<"Name: "<<name<<endl;
12     }
13 };
14 // Without Virtual
15 class EmployeeNV:public Person{
16     protected:
17         double salary;
18     public:
19     EmployeeNV(int id,string name,double salary):Person(id,name){
20         this-> salary = salary;
21     }
22     void displayEmployee(){
23         cout<<"Salary: "<<salary<<endl;
24     }
25 };
```

```
26 = class StudentNV: public Person{
27     protected:
28         double gpa;
29     public:
30 =         StudentNV(int id,string name,double gpa):Person(id,name){
31             this->gpa =gpa;
32         }
33 =         void displayStudent(){
34             cout<<"GPA: "<<gpa<<endl;
35         }
36 };
37 = class TANNV: public EmployeeNV,public StudentNV{
38     public:
39         TANNV(int id,string name,double salary,double gpa):EmployeeNV(id,name,salary),
40             StudentNV(id,name,gpa){}
41 =         void display(){
42             // displayPerson(); Ambiguous call
43             EmployeeNV::displayPerson();
44             // We are calling Display Person from a single class,so, No ambiguous call
45             displayStudent();
46             displayEmployee();
47         }
48 };
```

```
49 // with Virtual
50 = class Employee:virtual public Person{
51     protected:
52         double salary;
53     public:
54 =         Employee(int id,string name,double salary):Person(id,name){
55             this-> salary = salary;
56         }
57 =         void displayEmployee(){
58             cout<<"Salary: "<<salary<<endl;
59         }
60 };
61 = class Student:virtual public Person{
62     protected:
63         double gpa;
64     public:
65 =         Student(int id,string name,double gpa):Person(id,name){
66             this->gpa =gpa;
67         }
68 =         void displayStudent(){
69             cout<<"GPA: "<<gpa<<endl;
70         }
71 };
```

Yash Raj [25k-0878]

```
72 class TA: public Employee, public Student{
73     public:
74         TA(int id, string name, double salary, double gpa): Person(id, name), Employee(id, name, salary),
75         Student(id, name, gpa){}
76     void display(){
77         displayPerson();
78         displayStudent();
79         displayEmployee();
80     }
81 };
82 int main(){
83     cout<<"---Without Virtual---\n";
84     TANKV t1(2, "Yash", 23333, 1.2);
85     t1.display();
86     cout<<"---With Virtual---\n";
87     TA ta(1, "Yash", 20000, 2.9);
88     ta.display();
89 }
```

```
---Without Virtual---
Id: 2
Name: Yash
GPA: 1.2
Salary: 23333
---With Virtual---
Id: 1
Name: Yash
GPA: 2.9
Salary: 20000
```